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{
  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "09-07-2026_17:36:32",
  "variant" : 3,
  "subjectName" : "operating-systems",
  "categories" : [ "Deadlocks & Synchronization", "Memory Management", "Process Scheduling" ],
  "questions" : [ {
    "question" : "What condition occurs when multiple threads modify shared data concurrently with output depending on execution order?",
    "answers" : [ "Starvation anomaly", "Race Condition", "Deadlock state", "Livelock loop" ],
    "correctIndex" : 1
  }, {
    "question" : "What is a binary semaphore commonly called when used strictly as a locking mechanism?",
    "answers" : [ "Condition Variable", "Mutex", "Critical Section", "Monitor" ],
    "correctIndex" : 1
  }, {
    "question" : "What classical algorithm can be utilized to safely avoid deadlocks in resource allocation?",
    "answers" : [ "De Morgan's Optimization", "Banker's Algorithm", "Round Robin Algorithm", "Dijkstra's Shortest Path" ],
    "correctIndex" : 1
  }, {
    "question" : "What is the main difference between internal and external fragmentation?",
    "answers" : [ "External happens on hard drives; internal happens inside CPU registers", "Internal occurs inside fixed allocated blocks; external occurs between variable blocks", "Internal is fixed by defragmentation; external requires virtual memory flushes", "External happens inside kernel space; internal happens in user space" ],
    "correctIndex" : 1
  }, {
    "question" : "What state does a system enter when it spends more time swapping pages than executing instructions?",
    "answers" : [ "Paging", "Thrashing", "Starvation", "Fragmenting" ],
    "correctIndex" : 1
  }, {
    "question" : "What is the function of the Memory Management Unit (MMU)?",
    "answers" : [ "To partition hard drive sectors dynamically at runtime", "To clear cache registers when a context switch occurs", "To allocate network sockets to background daemons", "To map virtual addresses generated by the CPU to physical memory locations" ],
    "correctIndex" : 3
  }, {
    "question" : "What differentiates a process from a thread?",
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"answers" : [ "A single thread can contain multiple independent processes", "Processes share file descriptors by default; threads do not", "Threads run in kernel mode; processes run in user mode", "A process has its own isolated address space; threads share address spaces" ],
"correctIndex" : 3
}, {
"question" : "What state does a process enter if it is waiting for an I/O operation to finish?",
"answers" : [ "Zombie state", "Blocked / Waiting state", "Ready state", "Running state" ],
"correctIndex" : 1
}, {
"question" : "Which scheduling algorithm runs processes based on a fixed time-slice quantum rotation?",
"answers" : [ "First-Come, First-Served (FCFS)", "Multilevel Feedback Queue", "Round Robin (RR)", "Shortest Job First (SJF)" ],
"correctIndex" : 2
}, {
"question" : "What information is tracked inside a Process Control Block (PCB)?",
"answers" : [ "Process state, program counter, CPU registers, and memory boundaries", "Network interface card MAC address parameters", "The user web browser cache profiles", "Local database storage tables metadata" ],
"correctIndex" : 0
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